PROJECT REPORT

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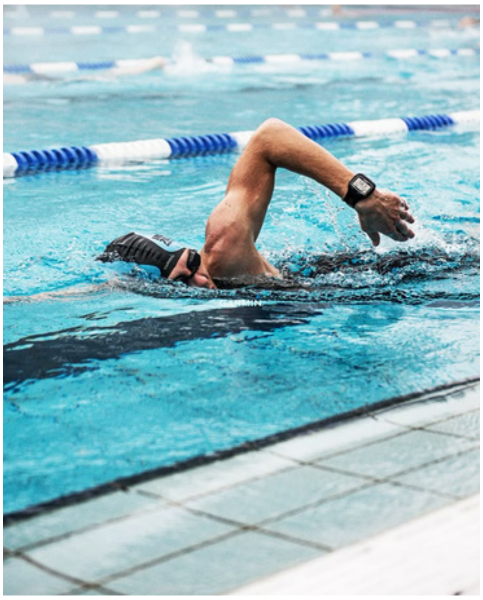
INTRODUCTION

GENERAL CONTEXT



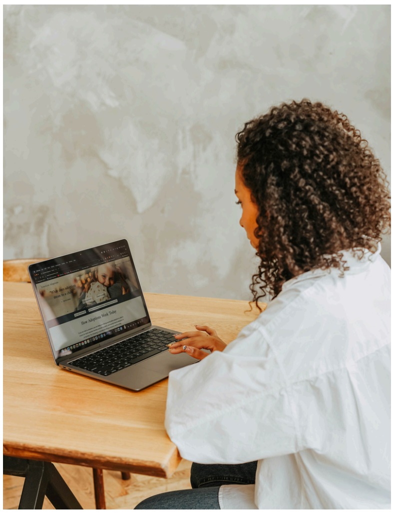
Swimming in Tunisia is a dynamic and competitive sport, involving athletes, coaches, clubs, and federations. However, the management of training programs, performance data, and communication is often fragmented and inefficient. There is a growing need for digital tools to enhance organization, analysis, and interaction in the swimming ecosystem.

PROJECT OBJECTIVES

The primary goal of the SWIFT project is to revolutionize the swimming experience in Tunisia by centralizing essential services into one unified platform. SWIFT aims to empower each stakeholder—swimmers, coaches, clubs, and federations—with smart tools and real-time information to help them achieve excellence.

RELEVANCE AND SCOPE

The primary goal of the SWIFT project is to revolutionize the swimming experience in Tunisia by centralizing essential services into one unified platform. SWIFT aims to empower each stakeholder—swimmers, coaches, clubs, and federations—with smart tools and real-time information to help them achieve excellence.



PROBLEM STATEMENT

REAL-WORLD ISSUE

Currently, training records, attendance monitoring, and performance analysis in Tunisian swimming are managed manually or using scattered tools, which leads to data loss, miscommunication, and inefficiencies in planning and performance tracking.

LIMITATIONS AND CHALLENGES

- Administrative Burden for Coaches and Clubs :

Coaches and club staff often rely on paper records or Excel sheets for attendance and performance tracking, which is time-consuming and prone to error.

- Lack of Centralized Communication :

Communication between coaches, swimmers, and administrators is often fragmented (phone calls, WhatsApp, emails), leading to misunderstandings and loss of important information.

- No Integration Between Training and Competition Data :

There's often no unified view that connects daily training data with actual competition performance, missing a critical feedback loop.

JUSTIFICATION FOR A DIGITAL SOLUTION

A digital solution is essential to streamline operations, enhance decision-making, and foster engagement. With SWIFT, users benefit from a connected ecosystem that simplifies processes and supports continuous improvement.

EXISTING SOLUTIONS

CURRENT TOOLS OR APPLICATIONS

There are several fitness and athlete-management apps available globally (e.g., MySwimPro, TeamUnify, Swim Manager). These tools are mostly generic and not tailored to the Tunisian context.

PROS AND CONS OF EXISTING SOLUTIONS

Advantages:

- Good user interfaces
- Global access to training plans
- Analytics features

Disadvantages:

- Lack of localization
- High cost for clubs/federations
- No integration with national performance metrics

HOW SWIFT STANDS OUT

SWIFT is customized for Tunisian users. It supports Arabic and French, integrates local federation goals, and includes a virtual assistant tailored to national competitions and training norms.

PROPOSED SOLUTION

This project is an innovative application designed to transform the experience of swimmers, coaches, and administrators within Tunisian swimming. It brings together in a single digital space:

- Personalized training programs for athletes,
- Management and attendance tracking tools for coaches,
- Real-time analytical dashboards for administrators,

And a virtual assistant (chatbot) to instantly answer users' questions.

Main Objective:

To create a unified community where each stakeholder : swimmer, coach, club, federation has access to clear information, effective tools, and smart support to aim for excellence.

Key Features:

- Intuitive interface to visualize one's journey and progress.
- Optimized session planning and management for coaches.
- Data analysis (results, rankings, reports) to support better decision-making at the federation level.
- Integrated chatbot for instant access to information (schedules, statistics, etc.).

SWIFT

NEEDS ANALYSIS

TARGET USERS

- Swimmers (athletes at all levels)
- Coaches and trainers
- Club administrators

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Functional Needs:

- Create and track training plans
- Monitor swimmer attendance
- Display performance dashboards
- Provide instant answers via chatbot

Non-Functional Needs:

- Security
- Responsiveness
- Usability and Accessibility
- Maintainability

DEVELOPED FEATURES

Below is a detailed description of the key features implemented in the SWIFT application:

FEATURE 1: COMPETITION DATA SCRAPING

Description:

- The system automatically scrapes competition results from official Tunisian swimming websites.

Objective:

- To centralize national competition data and make it easily accessible for analysis and tracking by federations and athletes.

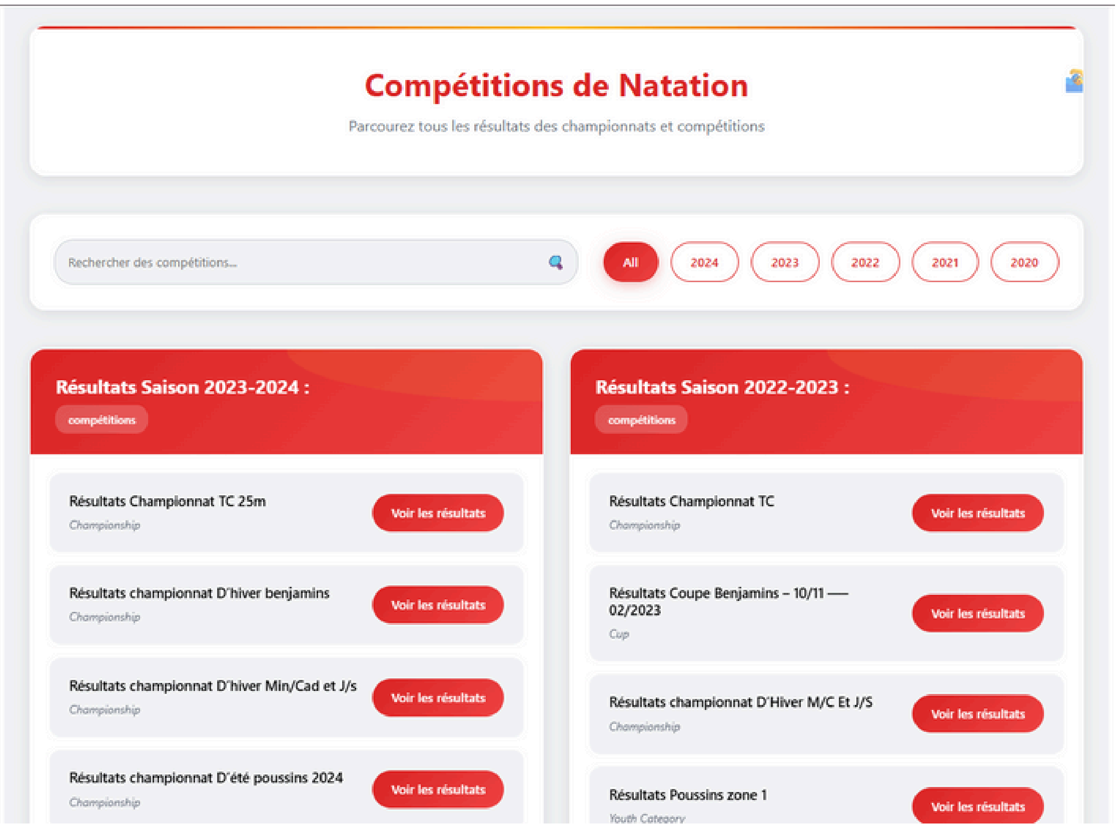


Figure 1.1 - swimming competitions

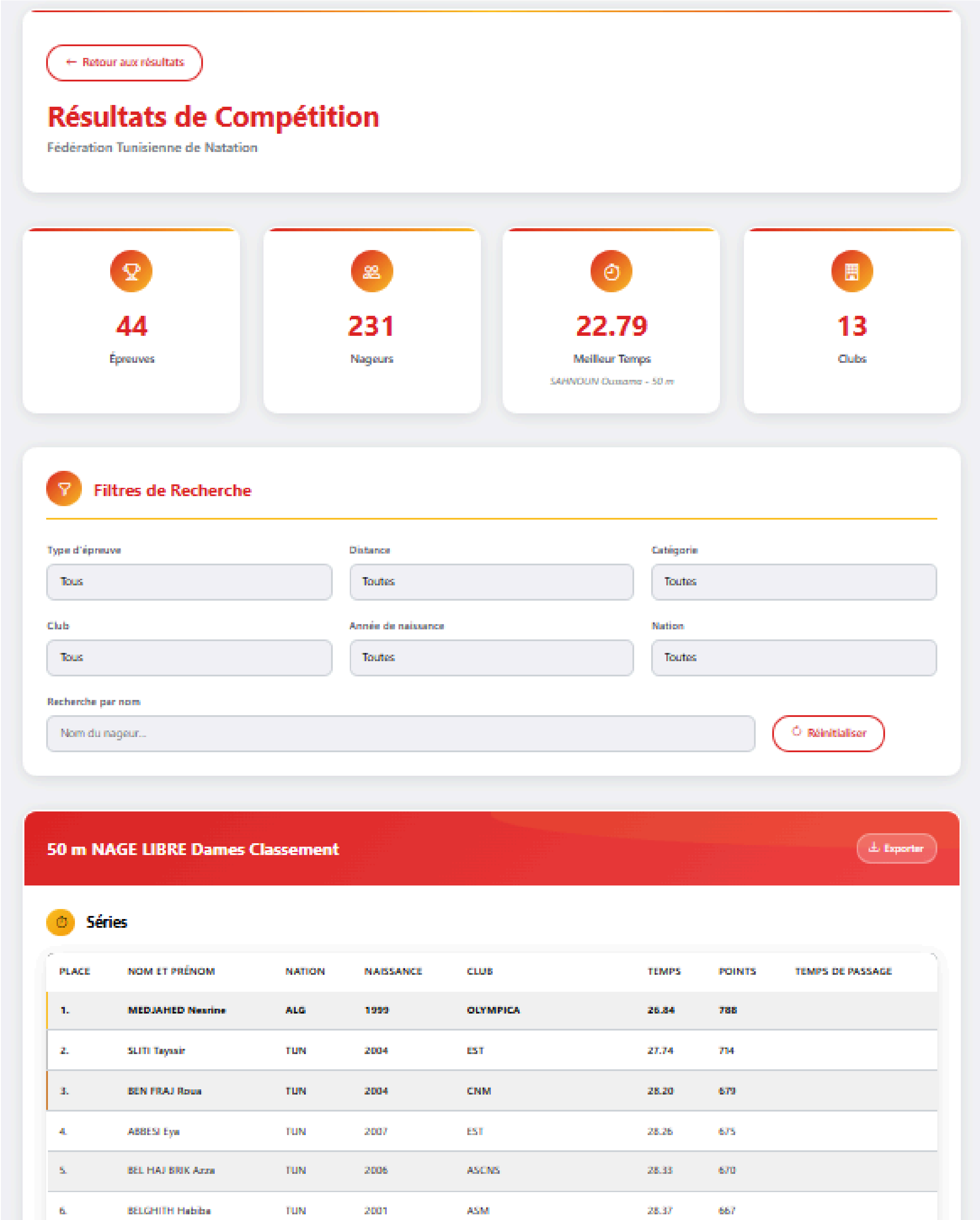


Figure 1.2 - competition results

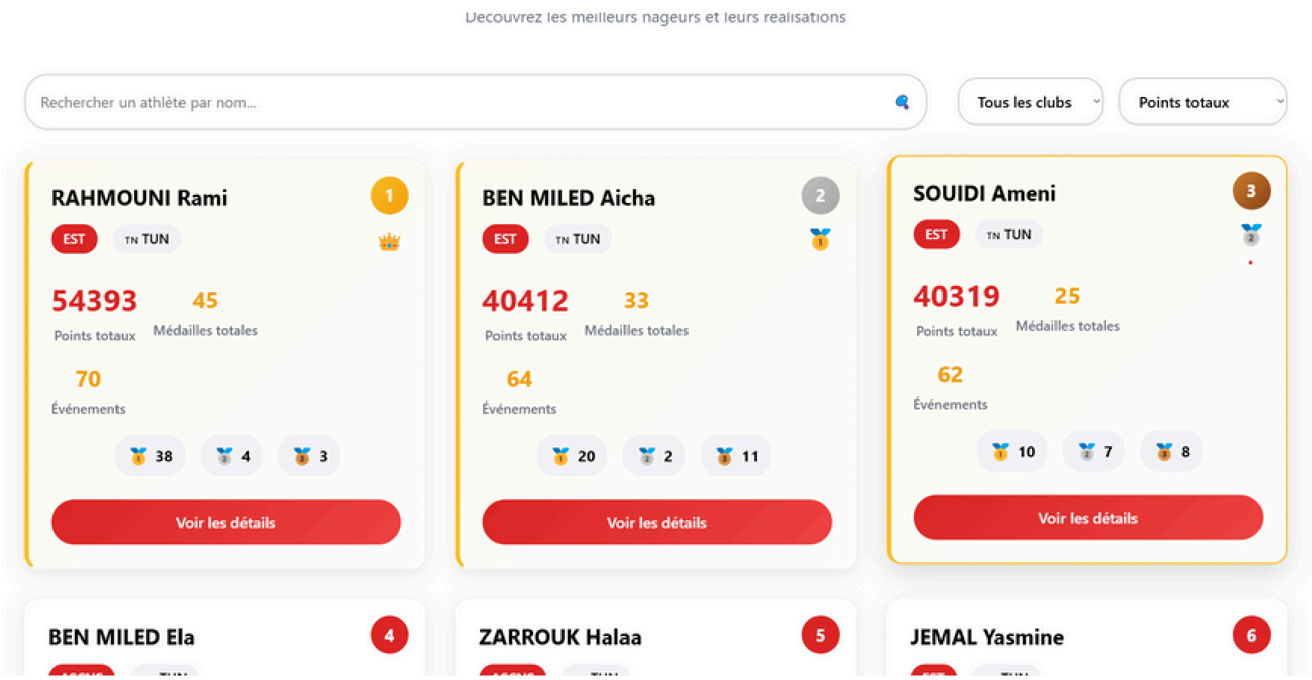


Figure 1.3 - athlete rankings

Figure 1.4 - global swimming statistics

FEATURE 2: ADMIN DASHBOARD

Description:

- A comprehensive admin interface for managing pools, training sessions, coaches, and swimmers.

Objective:

- To give the federation or club managers full control over the system's entities and visualize key statistics.

Submodules:

- Pool management (create/edit/delete)
- Training session planning
- Swimmer and coach administration
- Analytical dashboards for global insight

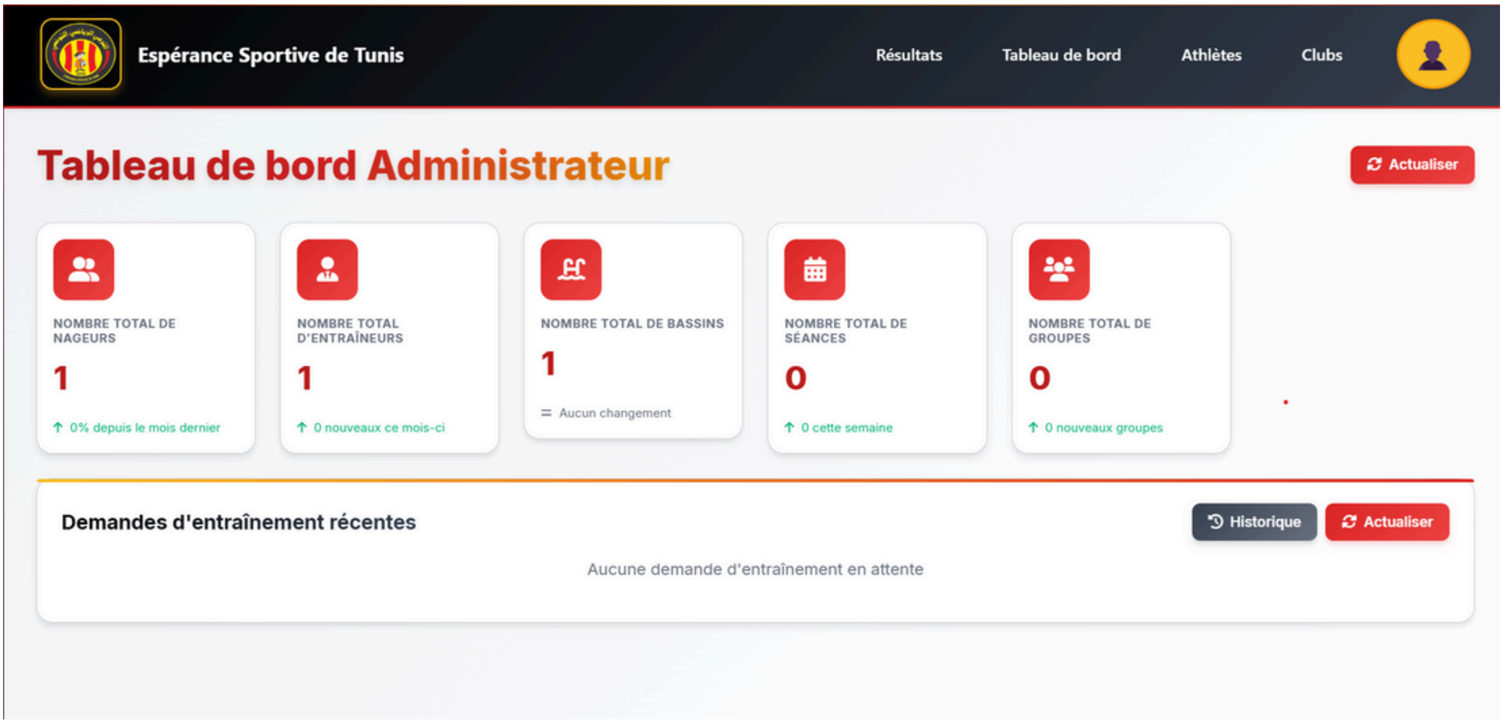


Figure 2 - admin dashboard

FEATURE 3: COACH DASHBOARD

Description:

- A personalized space for each coach to manage their profile, view and edit their training schedule, and track swimmer attendance.

Objective:

- To simplify coaches' daily routines and give them better tools for monitoring swimmer engagement.

Submodules:

- Profile update
- Weekly planning view and edits
- Attendance management (mark presence)

Figure 3.1 - Coach Dashboard

Figure 3.2 - training session schedule

Figure 3.3 - Attendance management (mark presence)

FEATURE 4: SWIMMER DASHBOARD

Description:

- A dedicated user space for swimmers to view their personalized schedules, track attendance, and update their profiles.

Objective:

- To improve swimmer engagement and allow them to be more autonomous in following their progress.

Submodules:

- Personal profile updates
- View weekly training plan
- Confirm or view attendance history

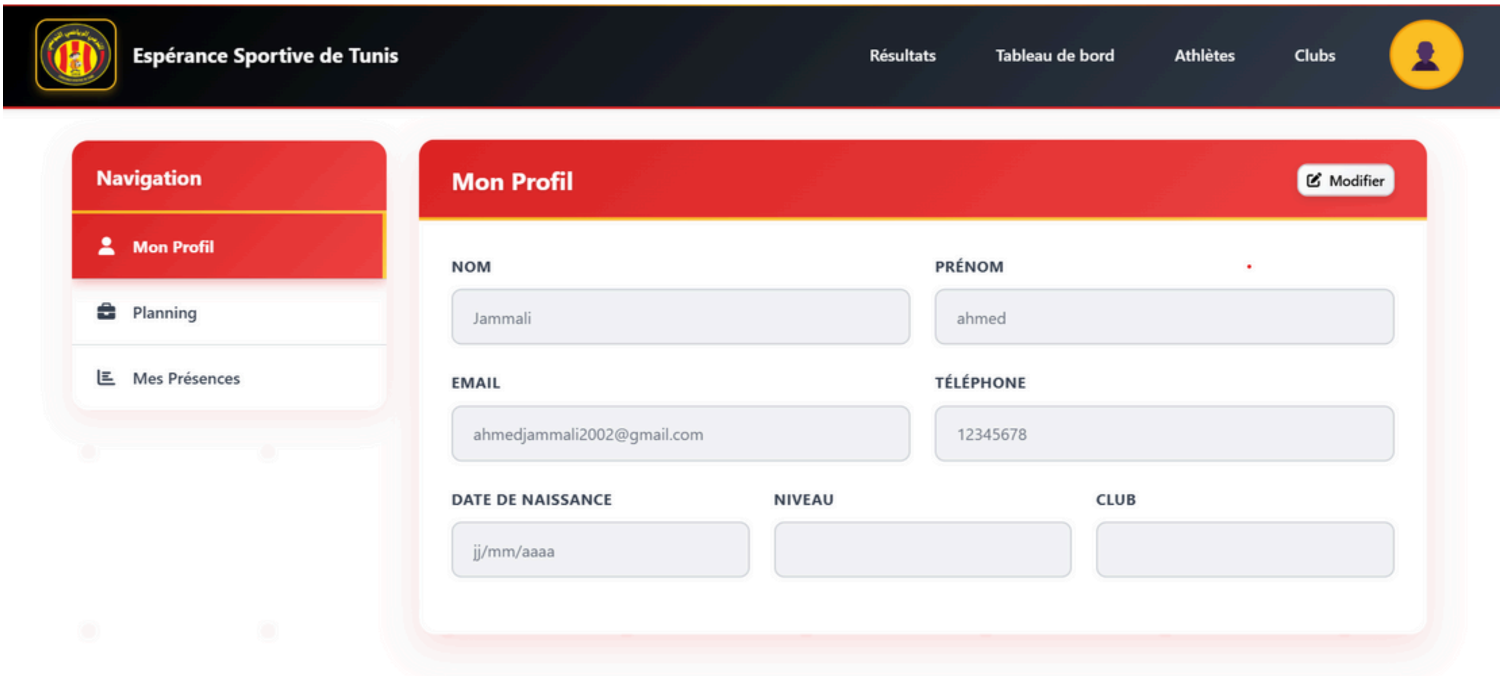


Figure 4.1 - athlete dashboard

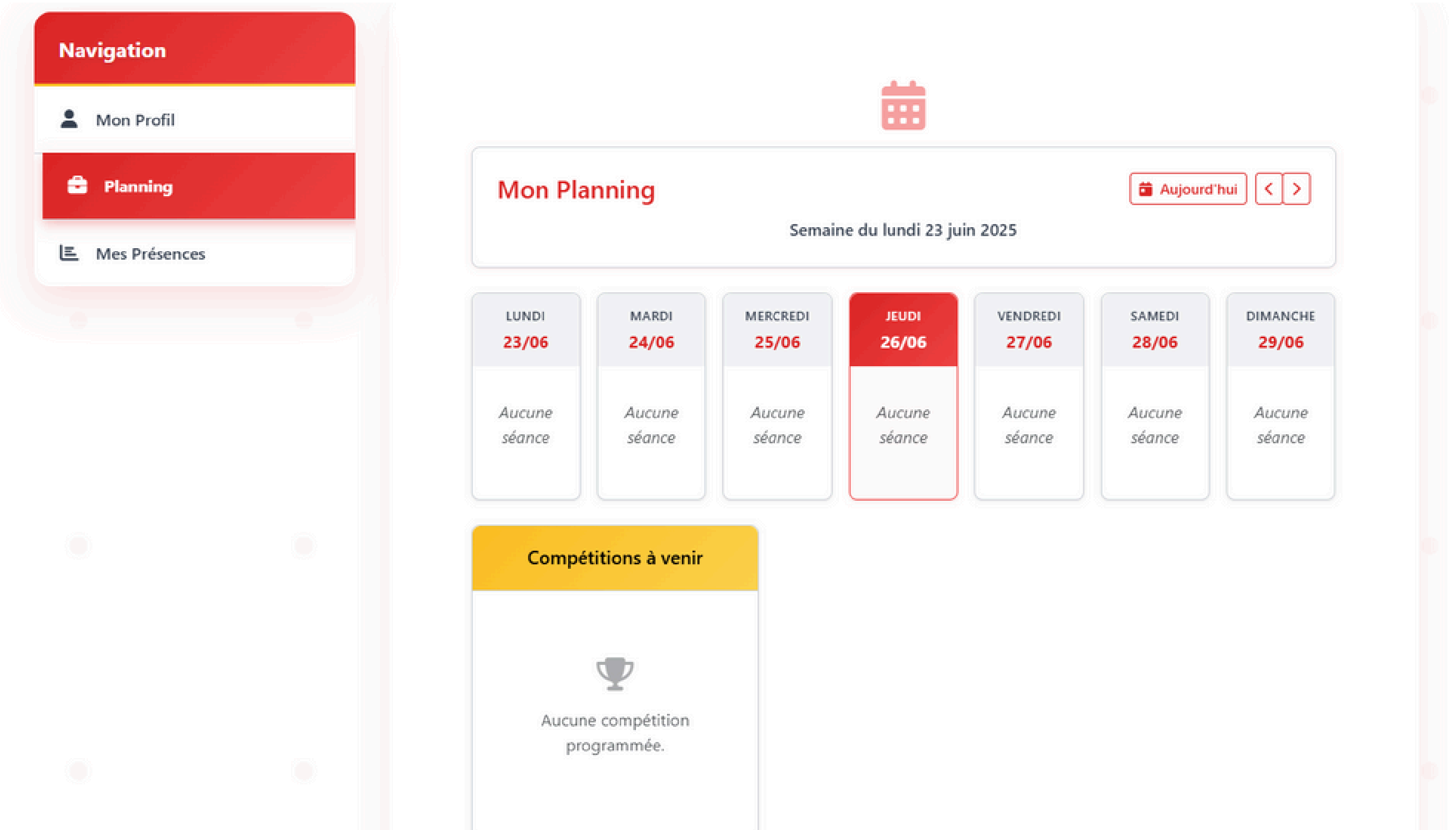


Figure 4.2 - training session schedule

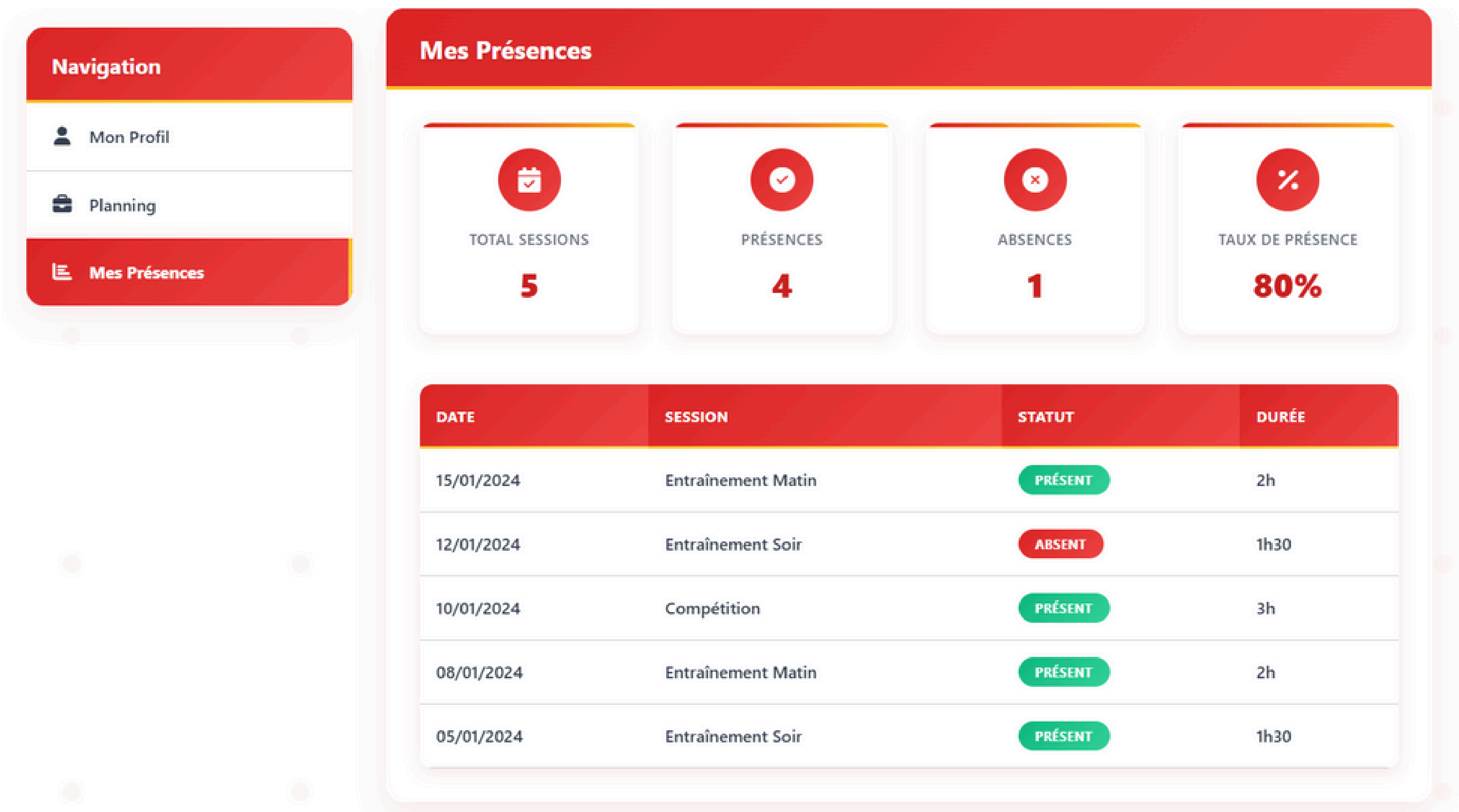


Figure 4.3 - attendance history

FEATURE 5: INTEGRATED CHATBOT

Description

- A virtual assistant that responds to users' questions about training sessions, schedules, competitions, and more.

Objective

- To provide quick, automated support and reduce the burden on administrative staff.

Key Features

- Responds instantly to questions about training schedules, competition results, and attendance
- Guides new users through platform navigation and available features
- Connects swimmers to training plans and performance data
- Offers basic nutrition diagnostics based on user input (age, training goals, diet habits)
- Available 24/7 with friendly and accessible language

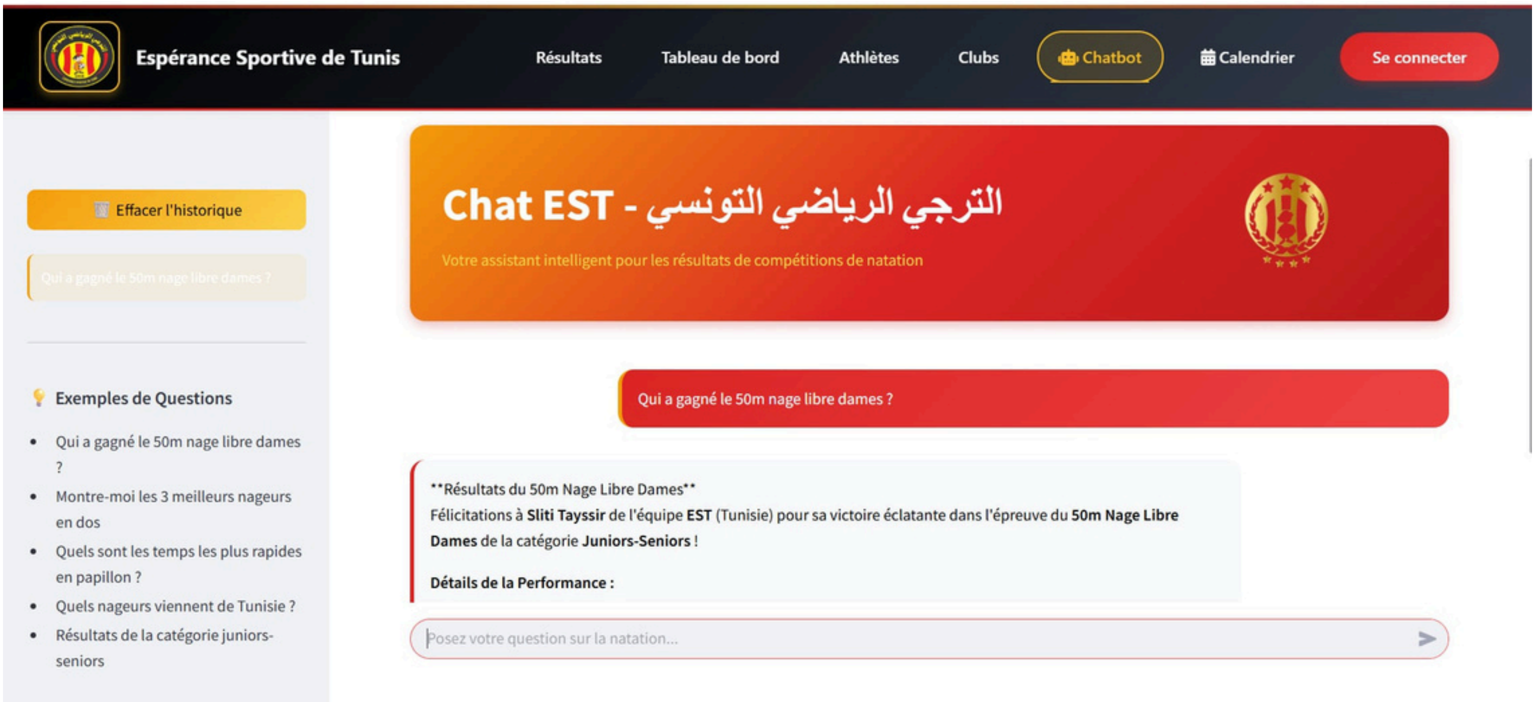


Figure 5.1 - Chatbot

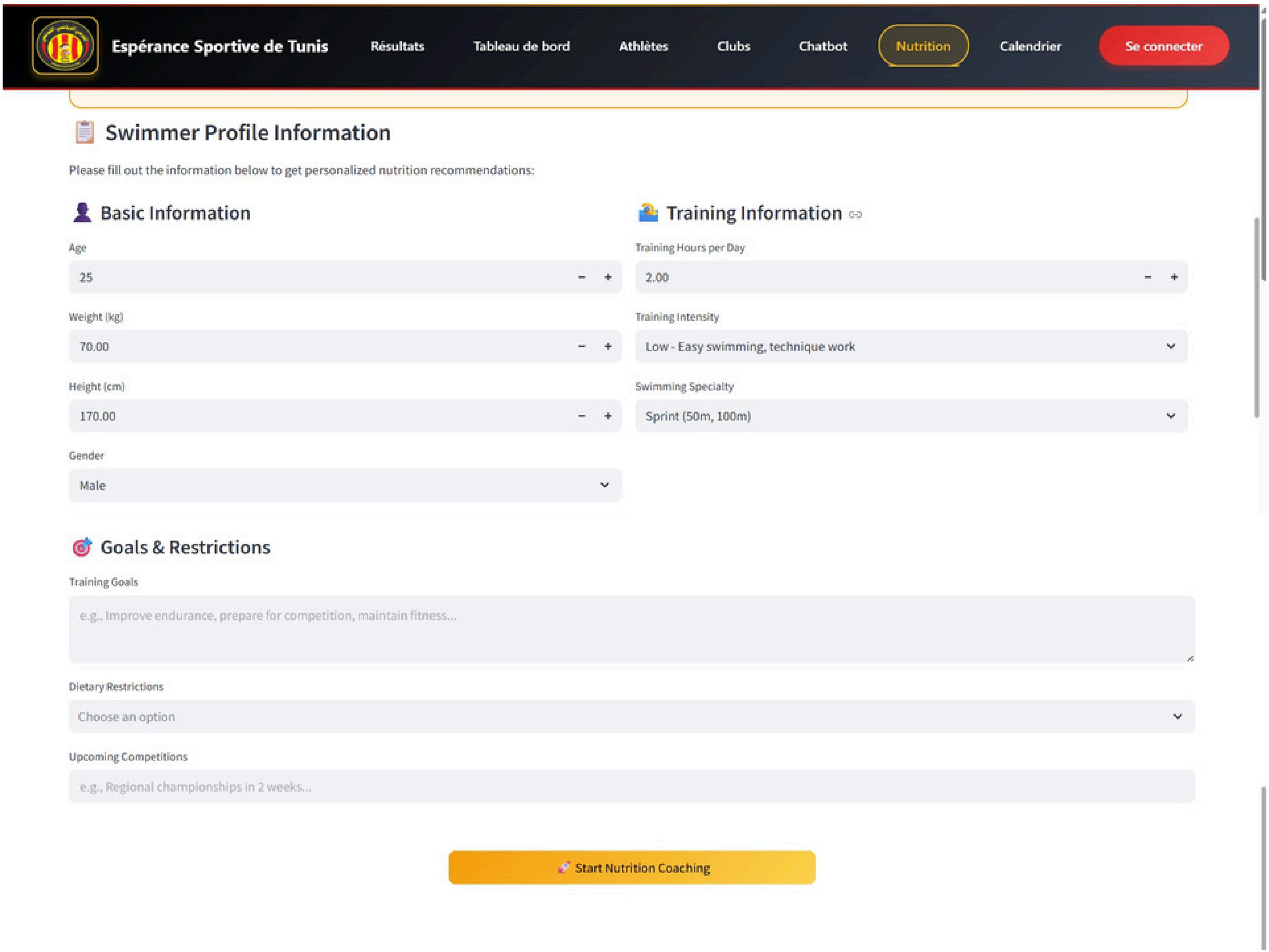


Figure 5.2 - AI-based nutrition coaching interface

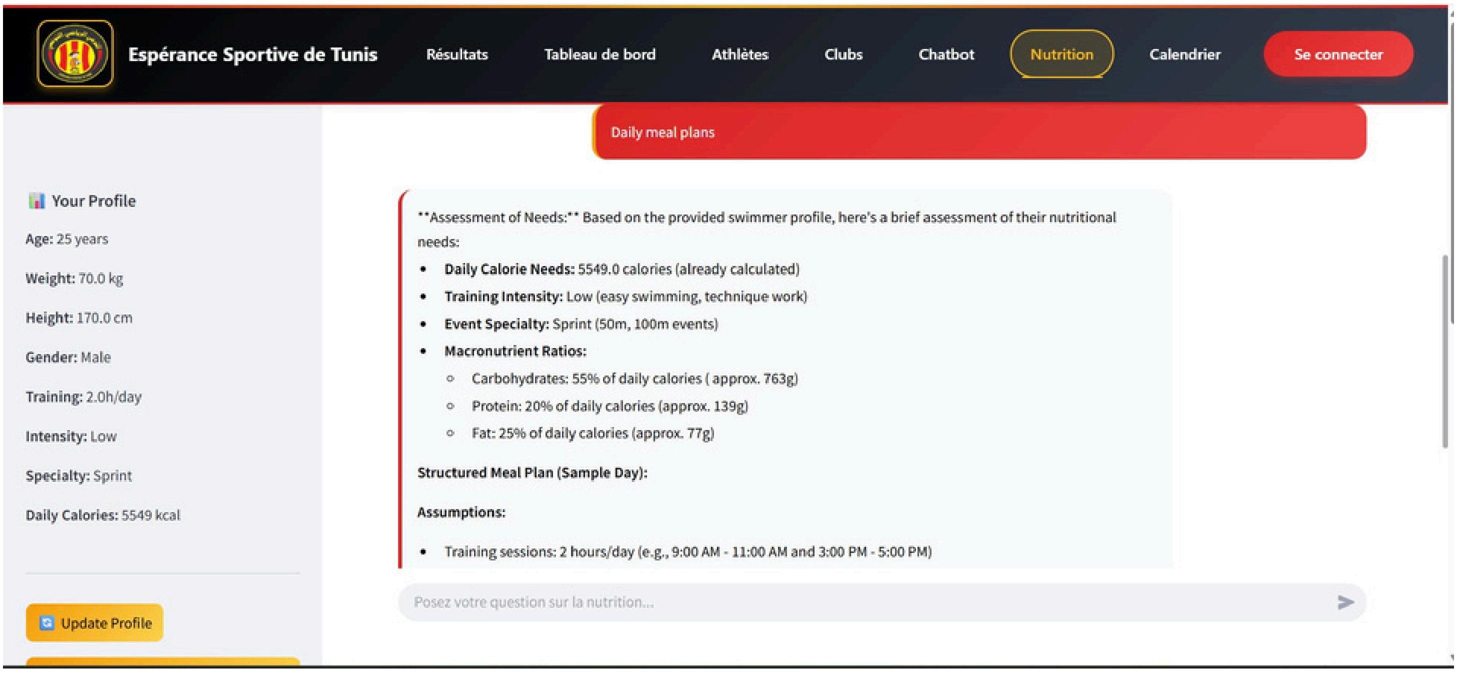


Figure 5.3 - AI-based Nutrition Diagnosis Interface

FEATURE 6: AUTHENTICATION & ACCESS CONTROL

Description:

- A secure authentication system that includes sign-up, login, and training session request submission.

Objective:

- To ensure secure access and proper role assignment for swimmers, coaches, and admins.

Features:

- Role-based login (swimmer, coach, admin)
- Sign-up with email validation
- Training request form (for swimmers wishing to join)

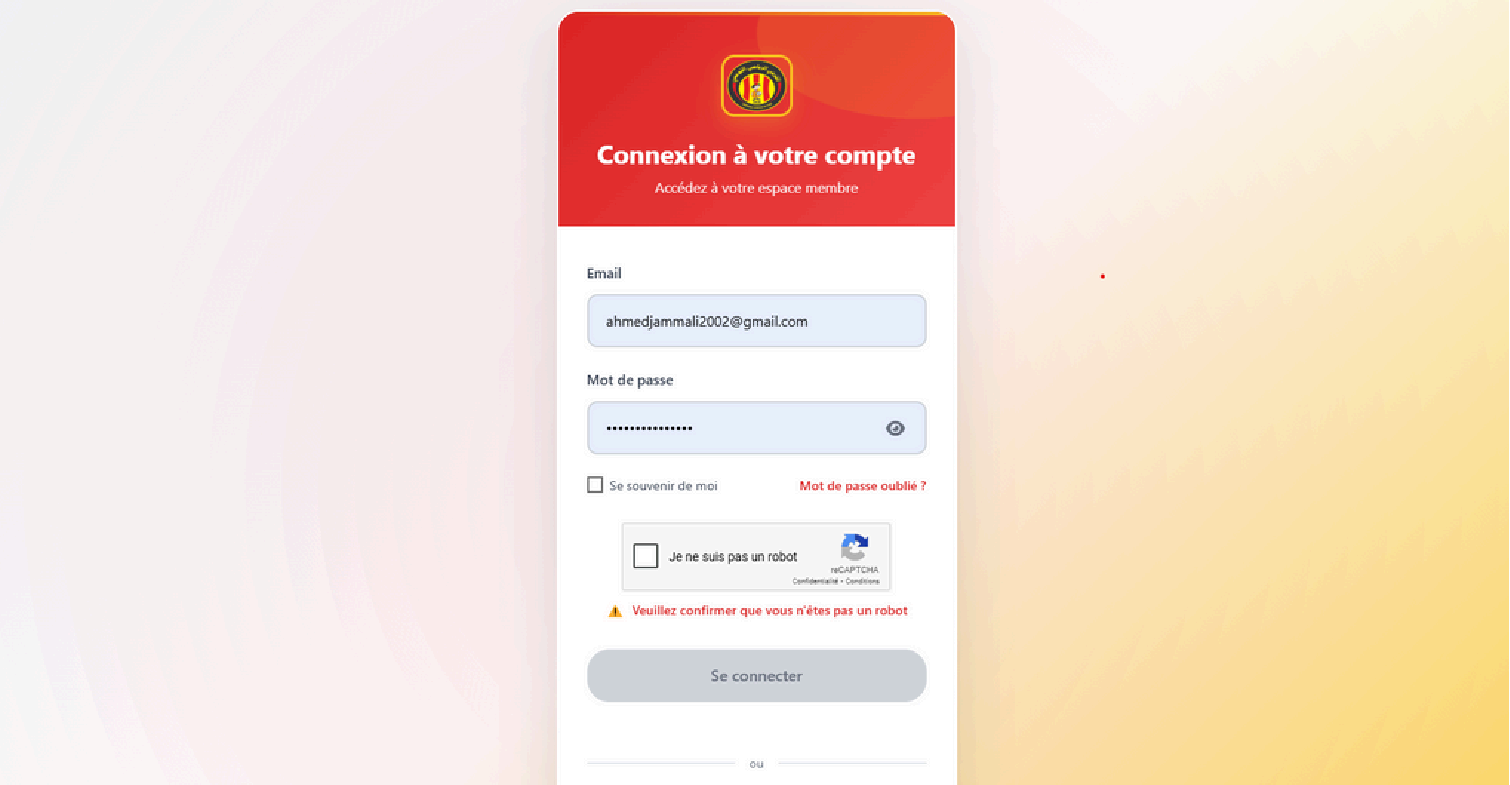


Figure 6.1 - Login Form

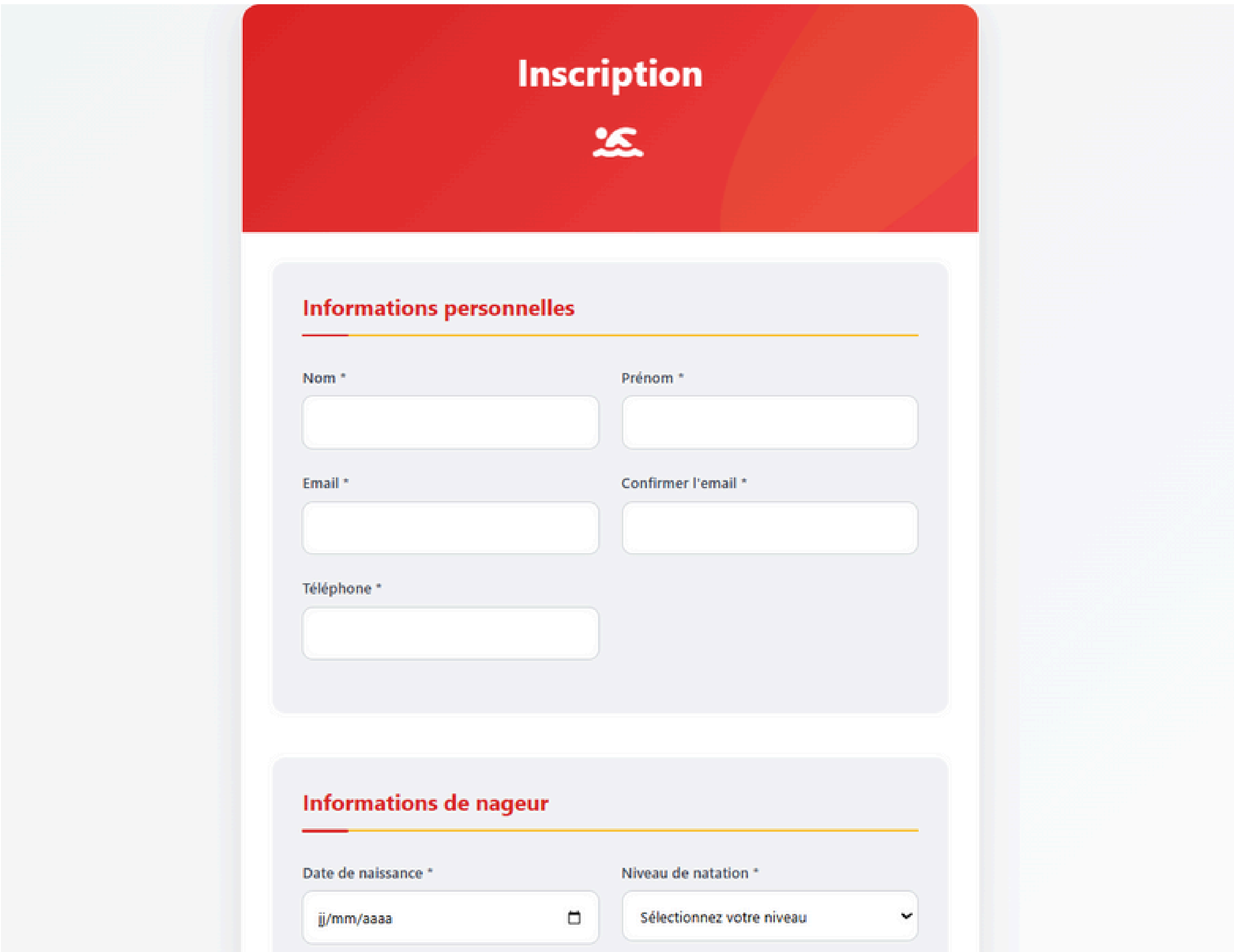


Figure 6.2 - Sign in Form

Figure 6.3 - Training request form

FEATURE 7: COMMUNITY FORUM

Description:

- An integrated discussion space where users can post questions, share experiences, and engage in conversations on various topics related to swimming.

Objective:

- To foster interaction among swimmers, coaches, administrators, and other community members. The forum centralizes discussions and strengthens collaboration within the SWIFT platform.

Features:

- Post questions by category (training, competition, equipment, etc.)
- Reply and comment in real-time
- Voting system to highlight the most helpful answers
- Moderation tools to manage inappropriate content
- Search and filter discussions by topic, date, or popularity

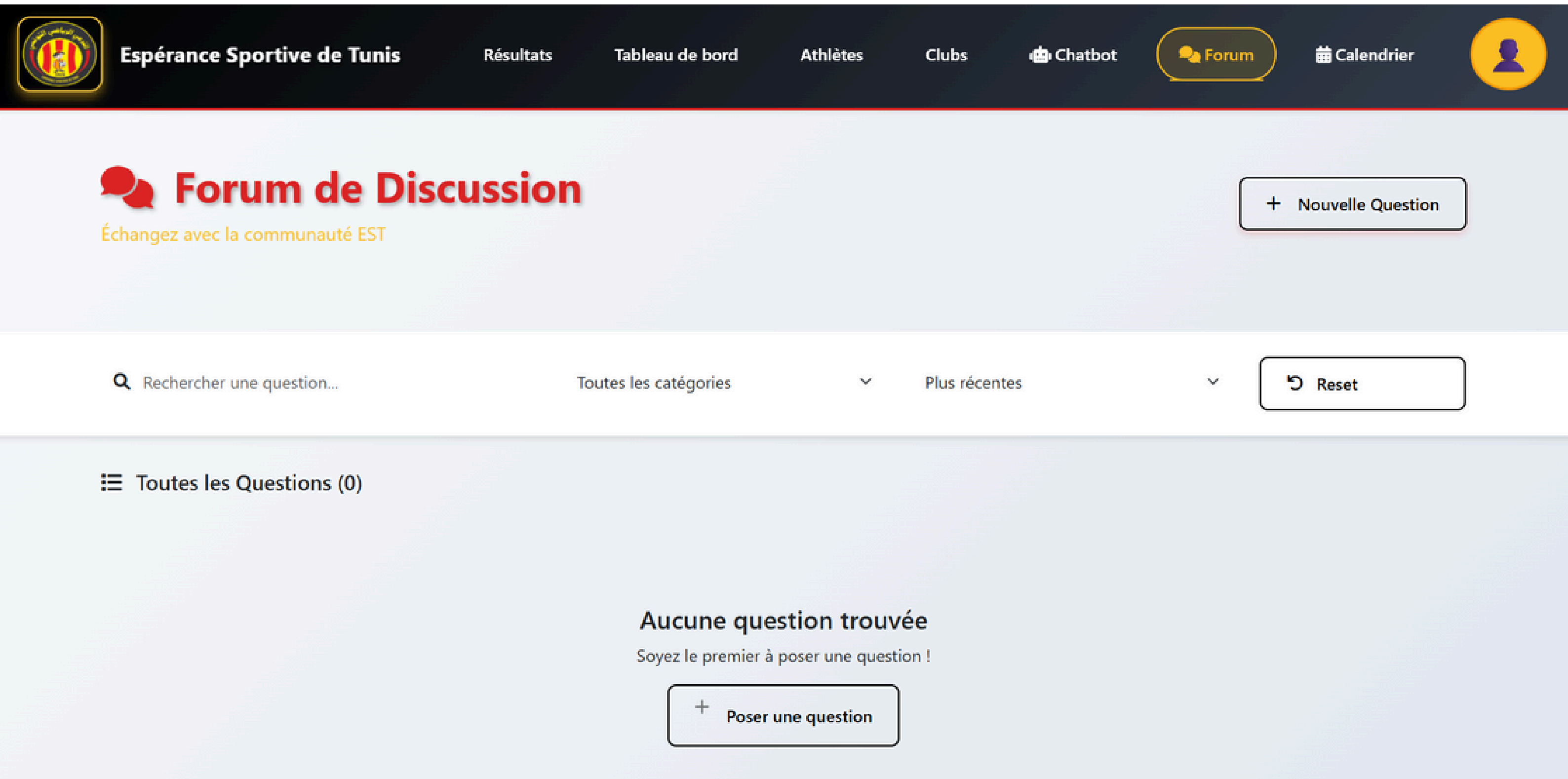


Figure 7 - Main forum page

TECHNICAL IMPLEMENTATION

TECHNOLOGIES USED

- Frontend: Angular 16
- Backend: Springboot
- Database: MySQL
- Chatbot: StreamLit / LangChain

TOOLS AND ENVIRONMENT

- Version control: GitHub / Git
- VSCode
- Intellij

CODE STRUCTURE

- Modular MVC architecture
- Reusable UI components
- API-driven backend

TECHNICAL CHALLENGES

- Real-time data synchronization
- Responsive UI adaptation
- Data Scraping Reliability
- Role-Based Access Control
- API Rate Limiting & Security
- Integration of Chatbot with Data

CONCLUSION

PROJECT SUMMARY

- SWIFT successfully introduces a modern digital solution to the Tunisian swimming community. It unifies training, tracking, management, and support into one intuitive platform.

STRENGTHS AND AREAS FOR IMPROVEMENT

Strengths:

- User-centered design
- Real-time features
- Strong potential for adoption by clubs and federations

To Improve:

- Expand chatbot features
- Add support for wearable devices
- Further performance testing on low-bandwidth networks

FUTURE PROSPECTS

- National-level adoption
- Expansion to other sports
- Integration with smart devices and AI-based performance prediction


THANK
YOU